

Maths Class:

Mathematics

Term 3 Examination

TIME ALLOWED: 2 hours

- Write your name and class at the top of this page.
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value and may be attempted.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.

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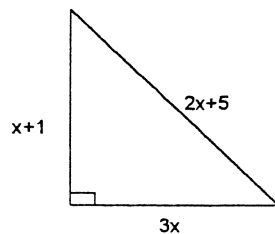
QUESTION 1: (11 MARKS)

- (a) Fully factorise:
- 1 (i) $2x^2 + 5x - 3$
- 2 (ii) $a^2 - b^2 - a + b$
- 1 (b) Simplify $3\sqrt{8} - \sqrt{32}$
- 1 (c) Find the value, to 2 dec places, of $\cos^2 32^\circ 11'$
- 3 (d) If $f(x) = x^2 + 3$, find
- (i) $f(-2)$ (ii) $\frac{f(x+h) - f(x)}{h}$
- 1 (e) Find the midpoint of the interval AB where A is (-2,3) and B is (6,-3)
- 2 (f) Rationalise the denominator of the following and simplify:

$$\frac{5\sqrt{3} - 1}{2 - \sqrt{3}}$$

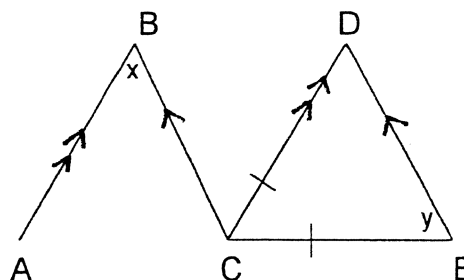
QUESTION 2: (11 MARKS)

- 2 (a) Solve and plot the solution on a number line to $6 - x \geq -3$
- 2 (b) Solve $(3x - 1)(2x + 5) = 0$
- 3 (c) Solve $2x^2 - 4x + 1 = 0$ giving your solutions correct to 2 decimal places.
- 4 (d) Form an equation to find the value of x in the following and then solve it to find x

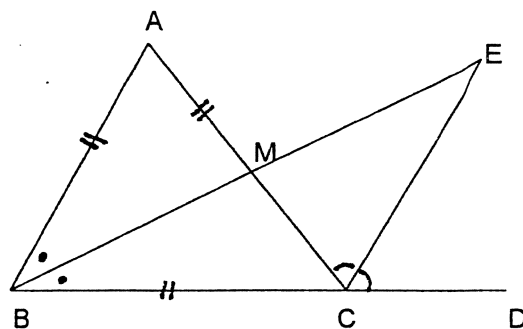


QUESTION 3: (11 Marks)

- 5 (a) In the diagram at right, $AB \parallel CD$ and $BC \parallel DE$. $\triangle CDE$ is isosceles. Showing all steps and giving all reasons, show that $x = y$.



- 6 (b) In the diagram at right, $\triangle ABC$ is equilateral. EB and EC bisect $\angle ABC$ and $\angle ACD$ respectively. Prove that $BC = CE$.



QUESTION 4: (11 MARKS)

- 2 (a) State whether the function $f(x) = x^2 - 2$ is odd, even or neither. Give reasoning.
- 2 (b) If $g(x) = \frac{1}{x^2} - 4$ find the value(s) of x for which $g(x) = 0$
- 2 (c) Sketch the graph of $y = \sqrt{16 - x^2}$ showing all important features.
- 4 (d) Shade the region given by the simultaneous solution of:
- $$\begin{cases} x^2 < y + 1 \\ y \leq 3 \\ x \geq 0 \end{cases}$$
- 1 (e) Give the natural domain of:

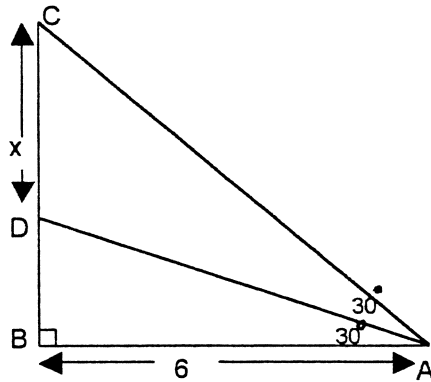
$$y = \frac{1}{\sqrt{3-x}}$$

QUESTION 5: (11 Marks)

- 3 (a) Find derivatives of:
- (i) $4x^5 - 2x^2 + 3$
- (ii) $\frac{5}{x^3}$
- (iii) $(3x-1)^4$
- 2 (b) Find the slope of the tangent to the curve $y = x^2(x-3)$ at the point (3,0)
- 3 (c) Find the equation of the tangent to the curve $y = 4 - x^2$ at $x = 1$
- 3 (d) Differentiate $y = \frac{3x^2}{x+5}$ and express your answer in simplest form

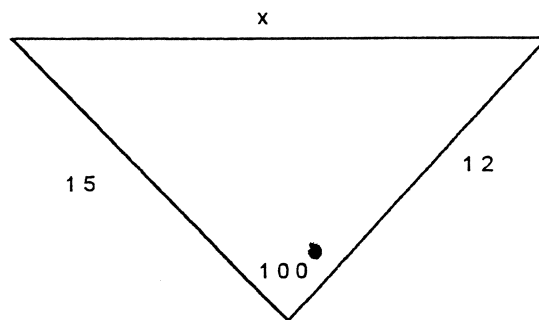
QUESTION 6: (11 marks)

3 (a)

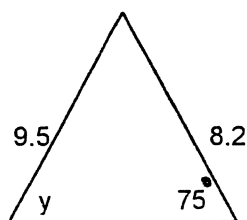


Show that $x = 4\sqrt{3}$

3 (b) Find the value of x to 2 decimal places in the following:



3 (c) In the following diagram, find the value of y to the nearest minute:



2 (d) If $\cos \theta = -5/13$ and $180^\circ < \theta < 360^\circ$, find $\tan \theta$ as a fraction.

QUESTION 7: (11 marks)

- 2 (a) The line $5x + ky = 4$ passes through the point $(-2, 1)$. Find the value of k .
- 3 (b) Find the equation of the line through $(-1, 4)$ and perpendicular to the line $3x + 4y = 5$
- 3 (c) Is the point $(5, 4)$ closer to the line $x + y = 2$ or is the origin closer? Give reasons.
- 3 (d) The points $A(0,0)$, $B(2,3)$, $C(3,5)$ and D complete a parallelogram.
- (i) Find the co-ordinates of the point D
- (ii) Find the length of the diagonal BD .

QUESTION 8: (11 Marks)

- 1 (a) Find an expression for $1 - \sin^2 \theta$
- 5 (b) For the parabola $y = 3x^2 + 5x - 2$ find $3x^2 - 12x + 9$
- (i) the intercepts on the x -axis
- (ii) the y -intercept
- (iii) the equation of the axis of symmetry
- (iv) the co-ordinates of the vertex
- 2 (c) Find $\frac{d}{dx}(3\sqrt{x})$
- 3 (d) Sketch the graph of $y = f(x)$ where $f(x)$ is defined by:

$$f(x) = \begin{cases} 1 & x < 0 \\ 1 - x & 0 \leq x \leq 2 \\ 3 - x^2 & x > 2 \end{cases}$$

SOLUTIONS

①

(a) (i) $(2n-1)(n+3)$ ①

(ii) $(a+b)(a-b) - (a-b)$ ①
 $= (a-b)(a+b-1)$ ①

(b) $6\sqrt{2} - 4\sqrt{2} = 2\sqrt{2}$ ①

(c) 0.72

(d) $f(n) = n^2 + 3$

(i) $f(-2) = 7$ ①

(ii) $\frac{f(n+h) - f(n)}{h} = \frac{(n+h)^2 + 3 - (n^2 + 3)}{h}$ ①
 $= \frac{n^2 + 2nh + h^2 + 3 - n^2 - 3}{h}$
 $= 2n + h$

(e) $M = (2, 0)$ ①

(f) $\frac{5\sqrt{3}-1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{10\sqrt{3}-2+15-\sqrt{3}}{1}$ ①
 $= 9\sqrt{3} + 13$ ①

③ (a) $\angle ABC = \angle BCD = x^\circ$ (alternate angles) ①
 $AB \parallel CD$

$\angle BCD = \angle CDE = x^\circ$ (alternate angles) ②
 $BC \parallel DE$

Because $\triangle CDE$ is isosceles, base angles are equal

$\therefore x = y$ ①

(b) Because $\triangle ABC$ is equilateral, ①
 all angles are 60°

$\therefore \angle MBC = \angle ABM = 30^\circ$ (BE bisects $\angle ABC$) ①

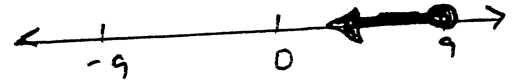
also $\angle ACD = 120^\circ$ (supplementary to $\angle BCM$) ①

$\therefore \angle MCE = \angle ECD = 60^\circ$ (CE bisects $\angle MCD$)

$\therefore \angle BCE = \angle BCM + \angle MCE = 120^\circ$

②

(a) $x \leq 9$



(b) $x = \frac{1}{3}$ or $x = -\frac{5}{2}$



(c) $x = \frac{4 \pm \sqrt{16-8}}{4}$
 $= \frac{2 \pm \sqrt{2}}{2}$ ②

(d) $(2n+5)^2 - (n+1)^2 + 9n^2$ ①

$4n^2 + 20n + 25$

$= n^2 + 2n + 1 + 9n^2$

$6n^2 - 18n - 24 = 0$

$n^2 - 3n - 4 = 0$

$(n-4)(n+1) = 0$

$n = 4$ or $n = -1$ ①

ONLY SOLUTION is $n = 4$ ①

$\therefore \angle MEC = 30^\circ$ (angle sum of $\triangle BCE$) ①

$\therefore$ Since $\angle MEC = \angle MBC = 30^\circ$
 $\triangle BCE$ is isosceles ①

$\therefore BC = CE$ ①

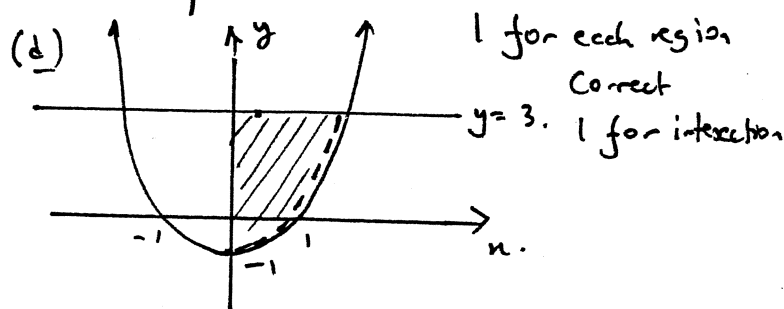
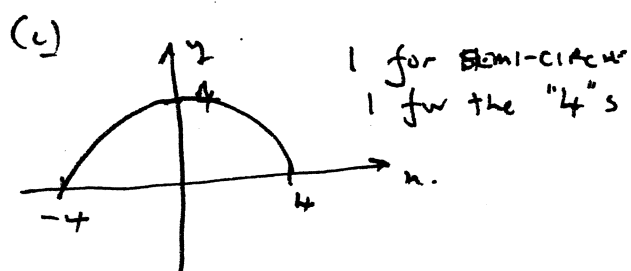
(4) (a) $f(a) = a^2 - 2$
 $f(-a) = (-a)^2 - 2$
 $= a^2 - 2$ } ①
 $\therefore f(a) = f(-a)$
 $\therefore$ EVEN FN. ①

(b) $g(n) = \frac{1}{n^2} - 4$

$\therefore \frac{1}{n^2} = 4$. ①

$\therefore n^2 = \frac{1}{4}$

$\therefore n = \pm \frac{1}{2}$ ①



(e) $n < 3$ ①
 (no mark for $n \leq 3$)

(6) (a) $\frac{BC}{6} = \tan 60^\circ$

$\therefore BC = 6\sqrt{3}$

$\frac{DB}{6} = \tan 30^\circ$

$\therefore DB = \frac{6}{\sqrt{3}} =$

$= \frac{6\sqrt{3}}{3}$

$= 2\sqrt{3}$

$\therefore n = 6\sqrt{3} - 2\sqrt{3}$

$= 4\sqrt{3}$

(5) (a) (i) $20n^4 - 4n$ ①

(ii) $-15/n^4$ ①

(iii) $12(3n-1)^3$ ①

(b) $y = x^3 - 3x^2$

$\frac{dy}{dx} = 3x^2 - 6x$ ①

At (3,0), $m_T = 27 - 18$
 $= 9$ ①

(c) $y = 4 - x^2$
 $\frac{dy}{dx} = -2x$ ①
 At $x=1$, $m_T = -2$. $y=3$ ①

Line is:

$y - 3 = -2(x - 1)$

$y - 3 = -2x + 2$

$2x + y - 5 = 0$ ①

(d) $y = \frac{3x^2}{x+5}$
 $\frac{dy}{dx} = \frac{(x+5)6x - 3x^2 \cdot 1}{(x+5)^2}$ ①

$= \frac{6x^2 + 30x - 3x^2}{(x+5)^2}$ ①

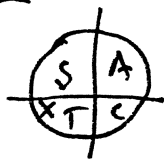
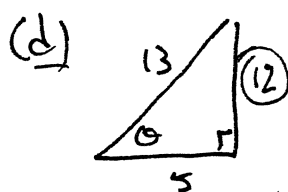
$= \frac{3x^2 + 30x}{(x+5)^2}$ ①

(b) $x^2 = 15^2 + 12^2 - 2 \cdot 15 \cdot 12 \cdot \cos 100^\circ$
 $x = 20.77$ (1) [1 off for not 2 dec pl.]

(c)
$$\left. \begin{aligned} \frac{\sin y}{8.2} &= \frac{\sin 75^\circ}{9.5} \\ \sin y &= \frac{8.2 \sin 75^\circ}{9.5} \end{aligned} \right\} (2)$$

$y = 56^\circ 29'$ (1)

[1 off for not nearest minute]



$\tan \theta = \frac{12}{5}$

(1) for true. (1)

(7)

(c) $5x + ky = 4$

Passes through $(-2, 1)$ (1)

$\therefore -10 + k = 4$

$k = 14$ (1)

(b) $3x + 4y = 5$

$y = -\frac{3x}{4} + \frac{5}{4}$

$\therefore \text{Slope} = -\frac{3}{4}$ (1)

$\therefore m_p = \frac{4}{3}$ (1)

Equation is:

$y - 4 = \frac{4}{3}(x + 1)$

$3y - 12 = 4x + 4$

$4x - 3y + 16 = 0$ (1)

(c) $p_1 = \frac{|1x_1 + my_1 + n|}{\sqrt{1^2 + m^2}}$

$= \frac{|1.5 + 1.4 - 2|}{\sqrt{1+1}}$

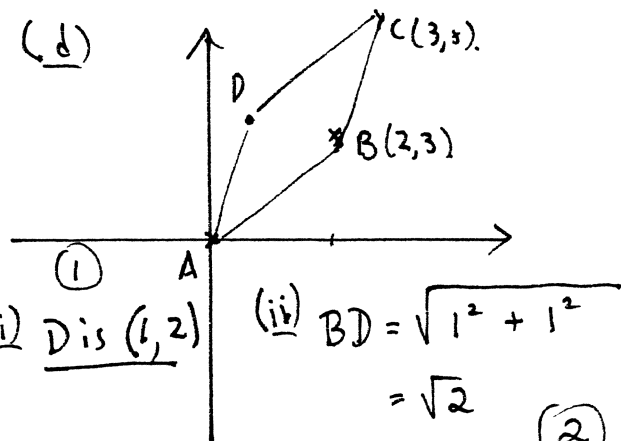
$= \frac{7}{\sqrt{2}} = \frac{7\sqrt{2}}{2}$ (1)

$p_2 = \frac{|1.0 + 1.0 - 2|}{\sqrt{2}}$

$= \frac{2}{\sqrt{2}} = \sqrt{2}$ (1)

$\therefore$ The origin is closer. (1)

(d)



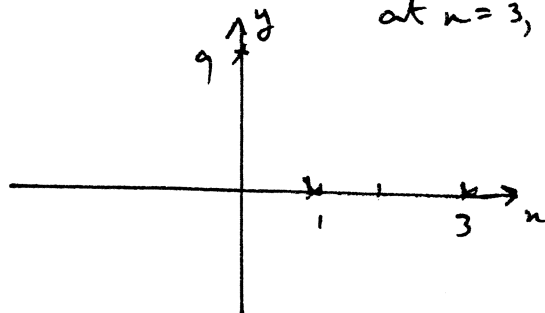
(i) Dis $(1, 2)$

(ii) $BD = \sqrt{1^2 + 1^2} = \sqrt{2}$ (2)

⑧ (a) $1 - \sin^2 \theta = \cos^2 \theta$

(b) $y = 3x^2 - 12x + 9$

(i) $y = 3(x^2 - 4x + 3)$
 $= 3(x-3)(x-1)$ ②
 at $x=3, x=1$

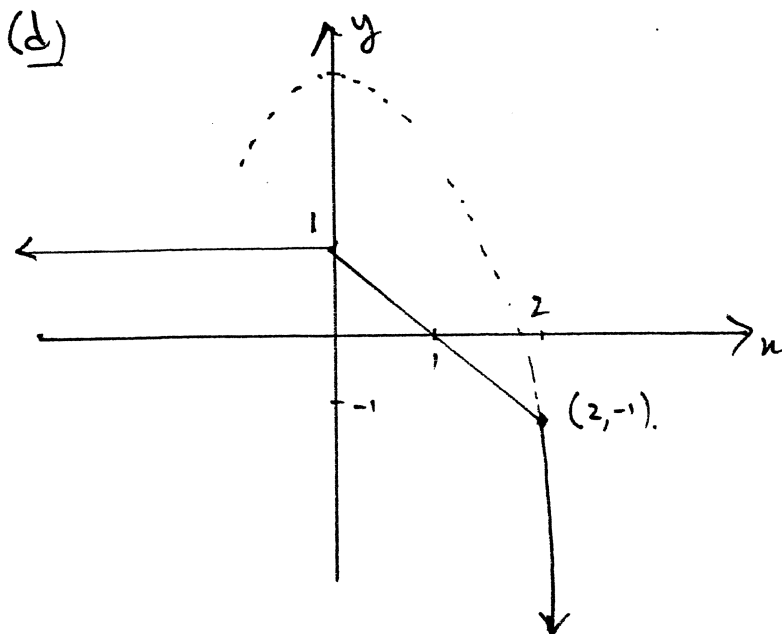


(ii) y-intercept is 9 ①
 or (0,9).

(iii) Axis is $x = 2$ ①

(iv) Vertex is (2, -3). ①

(c) $\frac{d}{dx}(3x^{1/2}) = \frac{3}{2}x^{-1/2}$ ①
 $= \frac{3}{2\sqrt{x}}$ ① ← not necessary.



3 marks

1 for each section
 of graph.
must indicate the
 point (2, -1)